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### DIFFERENCE BETWEEN COMPILER AND LINKER

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| FEATURE           | COMPILER                                   | LINKER                                       |
|-------------------|--------------------------------------------|----------------------------------------------|
| <b>PURPOSE</b>    | Translates high level code to machine code | Combines object file into a runnable program |
| <b>INPUT</b>      | Source code files                          | Object files, library                        |
| <b>OUTPUT</b>     | Object files (.obj, .o)                    | Executable files(.exe, .out)                 |
| <b>FUNCTIONS</b>  | Converts source code to machine code       | Resolves references, produces executable     |
| <b>CHECKS</b>     | Syntax, semantic, and type checking        | Resolves symbols and addresses               |
| <b>DEPENDENCY</b> | Each source file is compiled independently | Combine multiple files into one program      |
| <b>ERRORS</b>     | Syntax and semantic errors in source code  | Linking errors, unresolved references        |
| <b>STAGE</b>      | Single-stage process                       | Multi stage process (compile,link)           |

### For loop

A for loop in programming is a control flow statement that allows you to repeatedly execute a block of code based on a specified condition .

Example 1:

Basic increment loop:

```
#include<iostream>

using namespace std;

Int main(){
for (int i=0; i<5; i++) {
cout<<i<<" "
return 0;
}
}
```

Output:

0 1 2 3 4

[Example 2:](#)

Decrement loop:

```
#include<iostream>

Using namespace std;

Int main(){
for(int i = 5; i>0; i--)
cout<<i<<" ";
return 0;
}
}
```

Output:

5 4 3 2 1

[Example 3:](#)

Loop with custom step:

```
#include<iostream>

using namespace std;

int main(){
for (int i = 0; i<10 ; i+=2)
```

```
cout<<i<<" ";
return 0;
}
}
```

Output:

0 2 4 6 8

#### Example 4:

Nested loop:

Loops inside loop is called as nested loop

```
#include<iostream>
Using namespace std;
Int main(){
for(int i =0;i < 3;i ++)
for(int j =0;j<2; j++){
cout<<"("<i<<" ," << j << " )";
}
} return 0;
}
```

Output:

(0,0)(0,1)(1,0)(1,1)(2,0)(2,1)

#### Example 5:

Loop through array elements:

```
#include<iostream>
Using namespace std;
Int main(){
Int arr[] = {1 , 2 , 3 , 4 , 5};
```

```

For (int i = 0; i < 5, i++)
{
    cout<< arr[i] <<" ";
}
return 0;
}

```

Output:

1 2 3 4 5

## While loop

In while loop ,after initialization , condition is given and code with updation is written in body . Syntax with example is given below.

### Example 1:

Basic increment loop:

```

#include<iostream>

Using namespace std;

int main(){
    int i = 0;
    while(i < 5){
        cout<<i<<" ";
        i++;
    }
    return 0;
}

```

Output:

0 1 2 3 4

## Example 2:

Decrement loop:

```
#include<iostream>
```

```
using namespace std;
```

```
int main(){
```

```
int i =5;
```

```
while(i > 0);
```

```
cout<<i<<" ";
```

```
i--;
```

```
}
```

```
return 0;
```

```
}
```

Output:

5 4 3 2 1

## Example 3:

Loop with condition:

```
#include<iostream>
```

```
using namespace std;
```

```
int main(){
```

```
int value =10;
```

```
while(value>=0){
```

```
cout<<value<<" ";
```

```
value--;
```

```
}
```

```
return 0;
```

```
}
```

OUTPUT:

10 9 8 7 6 5 4 3 2 1 0

## Example 4:

Loop with user input:

```
#include<iostream>

Using namespace std;

Int main(){

Int userinput;

cout<<"enter a number(0 to exit): ";

cin>>userinput;

while (userinput!=0){

cout<<"entered: another number ( 0 to exit): ";

cin>>userinput;

}

return 0;

}
```

Output:

```
enter a number (0 to exit): 7

entered: 7

enter another number (0 to exit): 3

enter another number (0 to exit): 0
```

## Example 5:

loop with break statement:

```
#include<iostream>

Using namespace std;

Int main(){

Int I = 0;

while(true){
```

```
cout<< i<<" ";
```

```
l++;
```

```
if (l>= 5){
```

```
break;}
```

```
}
```

```
return 0;
```

```
}
```

Output:

0 1 2 3 4

\*\*\*\*\*END\*\*\*\*\*